

Identification and Control of a Propeller-Actuated Damped Pendulum (PADP)

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Abstract—This paper presents the experimental identification and closed-loop control of a propeller-actuated damped pendulum (PADP). Three step-response datasets were collected and processed using both manual parameter approximation and MATLAB’s System Identification toolbox, yielding six candidate transfer functions. Statistical analysis identified one experiment as disturbance-affected and excluded it; the final plant model was obtained by averaging the four remaining models. Three continuous-time controllers were designed — a PID tuned via IMC in SISOTool, an integral state feedback regulator via LQR (ISF-LQR), and an I-PD by pole placement — and physically validated against a settling time of 5 s and maximum overshoot of 3%. All three controllers satisfied both specifications experimentally; the ISF-LQR achieved the best transient performance ($t_s = 3.98$ s, $M_p = 0\%$), while the IMC-PID exhibited a pronounced control effort spike near steady state due to derivative action on the error signal.

I. INTRODUCTION

The propeller-actuated damped pendulum (PADP) is an underactuated mechanical system whose angular position is governed by the aerodynamic torque of a brushless motor and propeller assembly. Around a fixed operating point, the nonlinear dynamics admit a linear second-order underdamped approximation [1], making it a representative platform for comparing control strategies under the design requirements of settling time $t_s \leq 5$ s and maximum overshoot $M_p \leq 3\%$.

Plant identification was carried out using both parametric methods — extracting ω_n and ζ from measured transient features — and data-driven fitting via MATLAB’s *ident* toolbox [2]. Statistical screening across repeated trials was applied to handle noise and disturbances before model averaging.

Three control architectures were implemented and compared. A continuous PID tuned via Internal Model Control (IMC) [3], a Linear Quadratic Regulator with augmented integral state (ISF-LQR) [4], and an I-PD controller with pole placement [5], where the proportional and derivative terms act on the plant output to eliminate derivative kick and reduce overshoot.

This paper reports the complete identification-to-validation cycle for the PADP. Section II covers parameter estimation and plant derivation. Section III details the controller designs. Sections IV and V present and interpret the experimental results against the design specifications.

II. SYSTEM IDENTIFICATION

A. Experimental Setup and Data Acquisition

The PADP platform consists of an extruded-aluminium support rigidly fixed to a wooden base. A hollow aluminium arm pivots on bearings at the top of the support and carries a Hobbymate 2204 brushless motor governed by a SimonK 20 A ESC fitted with a GEMFAN 5045BNR three-blade propeller. Angular position is measured by a USDigital E5 incremental encoder (3600 CPR) coupled to the pivot shaft and sampled every 20 ms with a gain $k = 2\pi/3600$ rad/count. PWM actuation uses OneShot125 mode at a 500 μ s period. Three independent experiments were conducted by applying a step input of 1.7 units from rest, establishing a steady-state operating point near $\theta_{eq} \approx 0.6$ rad ($\approx 34.4^\circ$), and recording the angular response for 40 s.

B. Parameter Approximation

The logarithmic decrement method [1] was applied to each trial, yielding:

$$\delta = \frac{\ln(y_0/y_i)}{i}, \quad \zeta = \frac{\delta}{\sqrt{(2\pi)^2 + \delta^2}}, \quad \omega_n = \frac{2\pi/T}{\sqrt{1 - \zeta^2}} \quad (1)$$

yielding three second-order transfer functions of the prototype (eq. 2). Table I summarizes the parameters obtained by both methods.

$$G_i(s) = K_i \frac{\omega_{n,i}^2}{s^2 + 2\zeta_i \omega_{n,i} s + \omega_{n,i}^2}, \quad i = 1, 2, 3 \quad (2)$$

TABLE I: Identified parameters: approximation vs. SYSID

Method	Trial	K	ζ	ω_n (rad/s)
Approx.	1	0.338	0.0320	3.9292
	2	0.340	0.0350	3.9294
	3	0.423	0.0330	3.9291
SYSID	1	0.346	0.0292	3.9225
	2	0.346	0.0295	3.9301
	3	0.426	0.0296	3.9091

C. Statistical Analysis and Model Selection

The six models were pooled and their parameters compared statistically. Models from trial 3 deviated by more than $2s$ from the group mean, consistent with an external disturbance, and were excluded. The final plant was obtained by averaging the remaining four models:

$$G(s) = \frac{5.287}{s^2 + 0.2481s + 15.43} \quad (3)$$

III. CONTROLLER DESIGN

Design specifications for all regulators: settling time $t_s \leq 5$ s (2% criterion) and maximum overshoot $M_p \leq 3\%$ to a 0.6 rad step reference, with zero steady-state error and disturbance rejection.

A. Continuous PID Controller (SISOtool)

A continuous-time PID controller was designed using the Internal Model Control (IMC) tuning methodology. The IMC parameter was deliberately tightened to achieve $\%OS \approx 0\%$ in simulation, since the plant already introduces an inherent overshoot that could accumulate and violate the 3% bound in real operating conditions. The controller incorporates a filtered derivative action to attenuate high-frequency noise, as given by (4).

$$PID(s) = K_p + \frac{K_i}{s} + K_d s \cdot \frac{N}{s + N} \quad (4)$$

Coefficient matching of the IMC-equivalent open-loop transfer function yielded:

where K_p is the proportional gain, K_i the integral gain, K_d the derivative gain and N the derivative filter coefficient. The tuning procedure yielded the following parameter values:

TABLE II: IMC-tuned PID controller parameters.

K_p	K_i	K_d	N
-0.634	2.000	0.343	3

The negative proportional gain is consistent with the sign convention of the identified plant, which exhibits an inherently inverting static gain.

The PID controller with their values can be written as:

$$C(s) = -0.634 + \frac{2}{s} + 0.343 \frac{3s}{s + 3} \quad (5)$$

B. Integral State Feedback Controller (ISF) via LQR

An Integral State Feedback (ISF) controller was designed using the Linear Quadratic Regulator (LQR) criterion. The method augments the plant with an integral state to ensure zero steady-state error and improve disturbance rejection.

The identified plant in controllable canonical form is:

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} -0.2481 & -15.4300 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \\ \mathbf{C} &= [0 \quad 5.2870], \quad \mathbf{D} = 0 \end{aligned} \quad (6)$$

The system is fully controllable (rank = 2). To include integral action, the augmented model is defined as:

$$\dot{\mathbf{x}}_I = \mathbf{r} - \mathbf{C}\mathbf{x} \quad (7)$$

$$\mathbf{A}_a = \begin{bmatrix} -0.2481 & -15.4300 & 0 \\ 1 & 0 & 0 \\ 0 & -5.2870 & 0 \end{bmatrix}, \quad \mathbf{B}_a = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad (8)$$

The augmented system is also controllable (rank = 3). Based on the specifications ($t_s \leq 5$ s, $M_p \leq 3\%$), the desired poles were selected as:

$$p_{1,2} = -2.8 \pm 3.1266j, \quad p_3 = -14.0 \quad (9)$$

The LQR weighting matrices were chosen using Bryson's rule:

$$\mathbf{Q} = \text{diag}\left(\frac{1}{9^2}, \frac{1}{7^2}, \frac{1}{0.45^2}\right), \quad R = \frac{1}{0.6^2} \quad (10)$$

The resulting gain vector $\hat{\mathbf{K}} = [k_1 \ k_2 \ K_i]$ is:

$$\hat{\mathbf{K}} = [0.7214 \quad 0.4370 \quad -1.3333] \quad (11)$$

which yields the closed-loop poles:

$$p_{1,2} = -0.2593 \pm 3.9454j, \quad p_3 = -0.4509 \quad (12)$$

C. I-PD Controller via Pole Placement

The I-PD controller [5] prevents derivative kick by confining proportional and derivative action to the plant output, with only the integrator acting on the error:

$$C_{I-PD}(s) = \frac{K_i}{s} - (K_p + K_d s) \quad (13)$$

Dominant poles were derived from the specifications and a third real pole placed at $5|\text{Re}(p_{1,2})|$ to satisfy the non-dominance criterion [1]. Stricter targets ($t_s \leq 2.9$ s, $M_p \leq 1\%$) were imposed to absorb implementation degradation:

$$p_{1,2} = -1.3793 \pm j0.4705, \quad p_3 = -6.8966$$

$$K_p = 1.0817, \quad K_i = 2.7704, \quad K_d = 1.7793, \quad N = 20$$

D. Discrete PID Controller (SISOtool)

A discrete-time PID controller was designed through manual pole-zero placement in the z -domain, without resorting to an automated tuning method such as IMC. The compensator structure was chosen to consist of two real poles and two complex conjugate zeros, which corresponds to the discrete equivalent of a PID action with filtered derivative. The design procedure consisted of iteratively adjusting the controller gain, the natural frequency associated with the complex zero pair, and the pole locations until the closed-loop step response satisfied the specifications.

The resulting discrete controller transfer function is:

$$C(z) = \frac{3.0189(z^2 - 1.95z + 0.9881)}{(z - 0.09)(z - 1)} \quad (14)$$

IV. RESULTS AND ANALYSIS

This section presents the results obtained in both simulation and physical implementation for each controller, evaluating compliance with the performance specifications.

A. Continuous PID Controller

The simulated response settled at $t_s = 5.01$ s with $M_p = 3.4\%$, marginally outside specifications, consistent with the IMC tuning operating near the stability boundary. The experimental response is shown in Fig. 1; metrics for both conditions are summarized in Table IV.

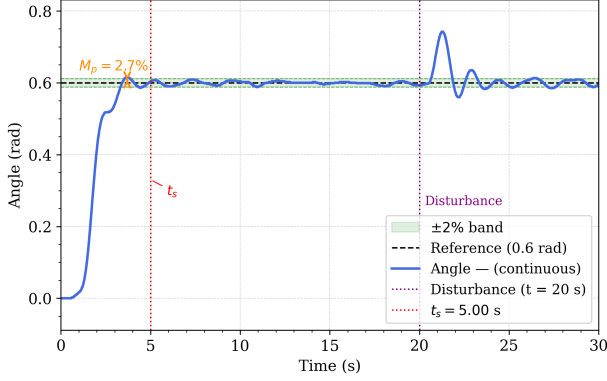


Fig. 1: Experimental step response of the PID controller.

The main simulation–experiment difference is the sustained oscillation inside the $\pm 2\%$ band in simulation, absent experimentally due to unmodelled high-frequency dynamics. The derivative filter was adjusted to $N = 5$ on the physical platform to attenuate these components, producing a smoother steady-state response than predicted by the linear model.

TABLE III: PID performance metrics: simulation vs. experiment

Condition	t_s (s)	M_p (%)	e_{ss}	Meets specs?
Simulation	5.01	3.4	0	No
Experimental	5	2.7	0	Yes

B. Integral State Feedback Controller via LQR

The linear simulation predicted $t_s = 9.94$ s due to the non-dominant pole $p_3 = -0.4509$ near the imaginary axis. The experimental response is shown in Fig. 2; both conditions are compared in Table V.

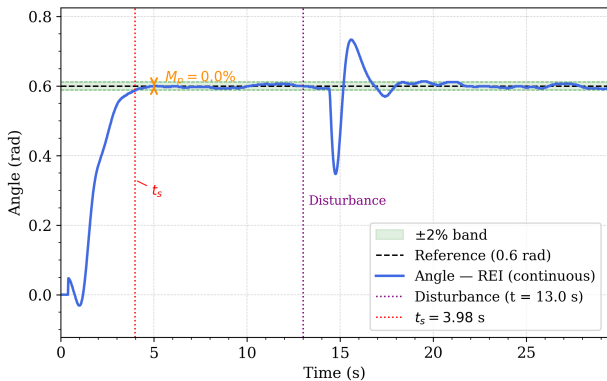


Fig. 2: Experimental step response of the ISF-LQR controller.

TABLE IV: ISF-LQR performance metrics: simulation vs. experiment

Condition	t_s (s)	M_p (%)	e_{ss}	Meets specs?
Simulation	9.94	0.12	0	No
Experimental	3.98	0	0	Yes

Both conditions satisfy $e_{ss} = 0$ and $M_p \leq 3\%$; the settling time discrepancy is addressed in Section V.

C. I-PD Controller via Pole Placement

The simulation with actuator saturation yielded $t_s = 3.9$ s and $M_p = 1\%$, consistent with the conservative pole placement ($\zeta_d \approx 0.946$). The mild oscillatory entry visible in Fig. 3 is caused by the motor saturating to zero when the pendulum exceeds the reference — a physically inevitable consequence of unidirectional thrust, not a design deficiency. Both conditions meet specifications (Table VI).

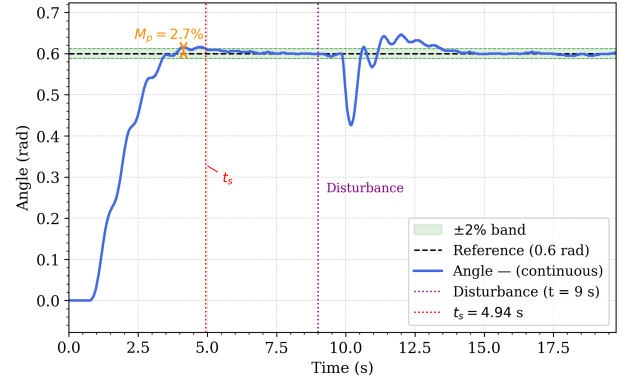


Fig. 3: Experimental step response of the I-PD controller.

TABLE V: I-PD performance metrics: simulation vs. experiment

Condition	t_s (s)	M_p (%)	e_{ss}	Meets specs?
Simulation	3.9	1.0	0	Yes
Experimental	4.94	2.7	0	Yes

D. Discrete PID Controller

The discrete PID controller (14), designed via manual pole-zero placement, failed to meet the imposed specifications on the physical plant. Unlike the continuous IMC design, where the tuning parameter provides a structured and systematic path to satisfy the closed-loop requirements, the manual approach is inherently sensitive to unmodelled dynamics. In discrete time, this sensitivity is further amplified by quantisation effects introduced by the sampling process, discretisation approximation errors in the derivative action, and the reduced phase margin that results from the computational delay added by the digital implementation. These factors, absent in the continuous design, cause the discrete controller to behave differently on the real platform than predicted by the linear model. Due to the unsatisfactory experimental outcome, no simulation or experimental plots are included for this controller.

E. Comparative Summary

Table VII consolidates the performance metrics of all three continuous-time controllers.

TABLE VI: Performance comparison: all controllers

Controller	Cond.	t_s (s)	M_p (%)	e_{ss}	Specs?
PID	Sim.	5.01	3.4	0	No
	Exp.	5.00	2.7	0	Yes
ISF-LQR	Sim.	9.94	0.12	0	No
	Exp.	3.98	0	0	Yes
I-PD	Sim.	3.9	1.0	0	Yes
	Exp.	4.94	2.7	0	Yes

V. DISCUSSION

A. ISF-LQR: Simulation–Experiment Discrepancy

Table IV shows a clear contrast between conditions. The slow simulated transient ($t_s = 9.94$ s) is dominated by the real pole $p_3 = -0.4509$ near the imaginary axis; experimentally, unmodelled nonlinear damping (friction, aerodynamic drag) suppresses this mode, yielding monotonic convergence at $t_s = 3.98$ s with zero overshoot. The initial negative deflection (≈ -0.07 rad at $t \approx 1$ s) reflects actuator dead-zone and propeller inertia absent from the linear model. Disturbance rejection is confirmed at $t \approx 13$ s, where a 0.25 rad excursion is fully recovered by the integral state, validating closed-loop robustness despite model inaccuracies.

B. Continuous PID via IMC

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C. I-PD via Pole Placement

The I-PD controller achieved the most consistent agreement between simulation and experiment among the three designs. The conservative pole placement ($\zeta_d \approx 0.946$), corresponding to stricter targets of $t_s \leq 2.9$ s and $M_p \leq 1\%$, provided sufficient margin to absorb the degradation introduced by actuator saturation and ESC dead-band during physical implementation, yielding experimental metrics of $t_s = 4.94$ s and $M_p = 2.7\%$.

The structural advantage of the I-PD over the IMC-PID is evident in the transient entry: by confining derivative and

proportional action to the plant output, the reference step does not excite the derivative channel, avoiding the control effort spike that contributes to overshoot in the parallel PID structure. The oscillatory entry visible in the experimental response (Fig. 3) is not a design deficiency but a direct consequence of the unidirectional propeller thrust saturating to zero when the pendulum exceeds the reference, a nonlinearity absent from the linear simulation.

Compared to the ISF-LQR, the I-PD offers a more transparent design path: pole placement provides a direct mapping from time-domain specifications to closed-loop pole locations, without requiring the selection of LQR weighting matrices $\mathbf{Q}$ and $\mathbf{R}$, whose physical interpretation is less intuitive. However, the ISF-LQR benefits from explicitly using the velocity state to damp oscillations, which explains its faster experimental settling ($t_s = 3.98$ s vs. 4.94 s) and zero overshoot even in the presence of nonlinear plant dynamics.

VI. CONCLUSIONS

This work completed the full identification-to-validation cycle for the PADP: experimental data acquisition, statistical model selection, controller design, and physical validation.

All three continuous-time controllers satisfied the specifications of $t_s \leq 5$ s and $M_p \leq 3\%$ experimentally. However, the IMC-PID exhibited a pronounced control effort spike when driving the steady-state error to zero, a consequence of the derivative action on the error signal amplifying the plant's resonant response — its linear simulation additionally operated at the stability boundary, marginally failing both metrics. The ISF-LQR achieved the best experimental performance ($t_s = 3.98$ s, $M_p = 0\%$) through active velocity-state damping and integral disturbance rejection. The I-PD offered the most transparent design path, with pole placement providing a direct mapping from specifications to closed-loop dynamics and structural immunity to derivative kick, yielding ($t_s = 4.94$ s, $M_p = 2.7\%$).

The consistent gap between simulation and experiment across all controllers reveals the primary challenge of this project: the averaged linear model does not capture the ESC dead-band, propeller thrust nonlinearity, and bearing friction present in the physical system. Future work should address this through nonlinear plant modeling, gain scheduling across the operating range, or an active disturbance rejection strategy to improve simulation fidelity and robustness beyond the linearization point.

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